

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-31

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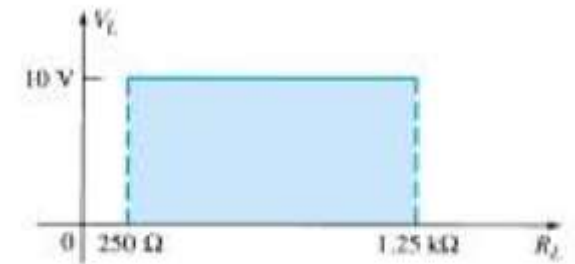
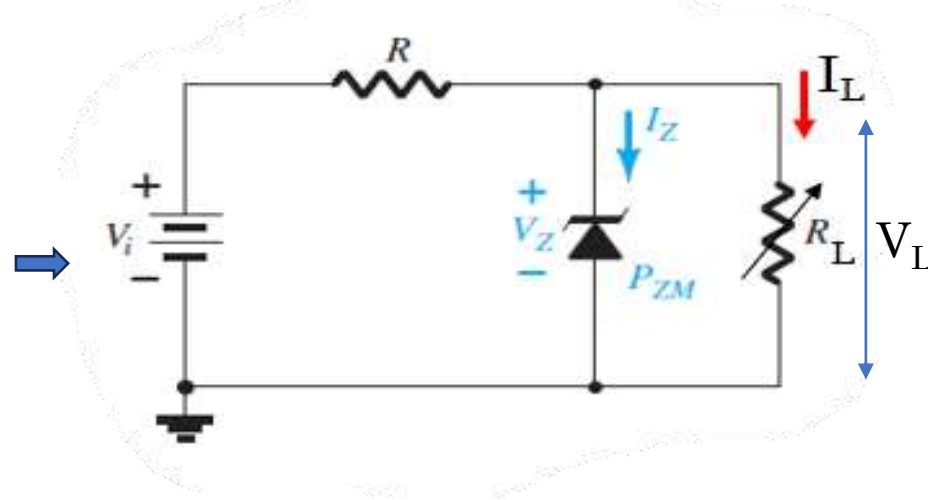
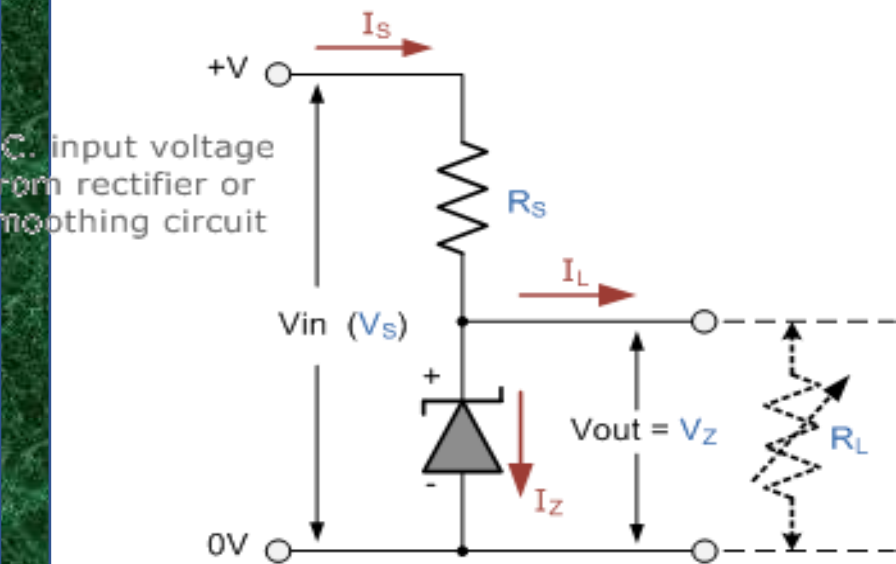
Outline of Today's Class

- Problems on Zener diode: case 2 (Variable R_L)

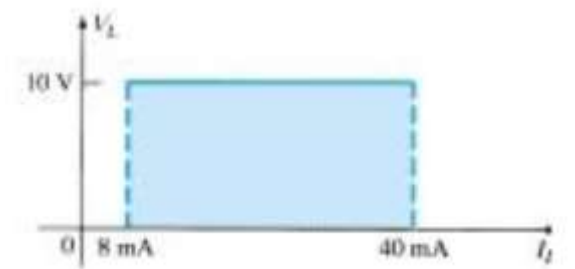
Case 2: Zener Regulator (Variable R_L)

Case 2: Zener Regulator (Variable R_L)

- Determine range of R_L and I_L that will result in V_{RL} being maintained at constant voltage of V_Z .



(a)



(b)

(a) R_L min.

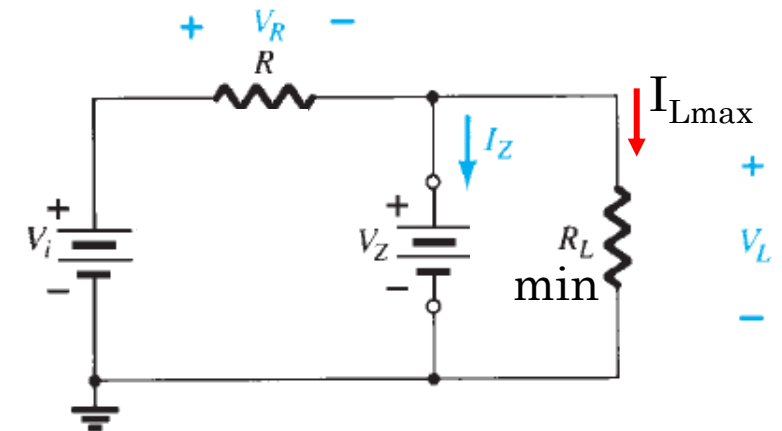
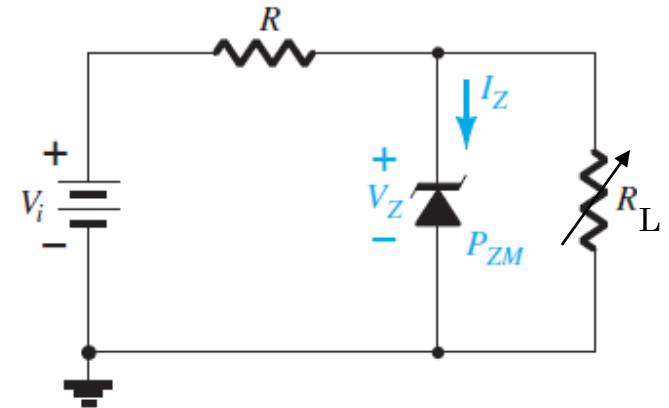
To determine the minimum load resistance of Fig. that will turn the Zener diode on, simply calculate the value of R_L that will result in a load voltage $V_L = V_Z$. That is,

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

- Minim voltage drop needs minimum resistance

Solving for R_L , we have

$$R_{L_{\min}} = \frac{RV_Z}{V_i - V_Z}$$



(b) I_L max.

The condition defined by
ifies the maximum I_L as

$$I_{L_{\max}} = \frac{V_L}{R_L} = \frac{V_Z}{R_{L_{\min}}}$$

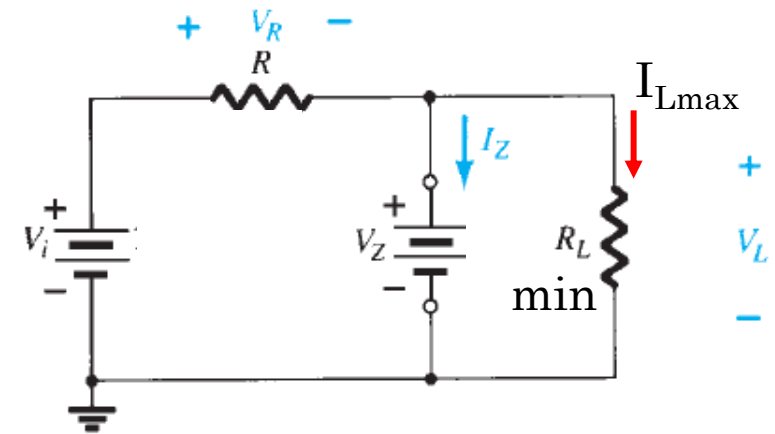
establishes the minimum R_L but in turn spec-

Once the diode is in the “on” state, the voltage across R remains fixed at

$$V_R = V_i - V_Z$$

and I_R remains fixed at

$$I_R = \frac{V_R}{R}$$



(c) I_L min and (d) R_L max.

The Zener current

$$I_Z = I_R - I_L$$

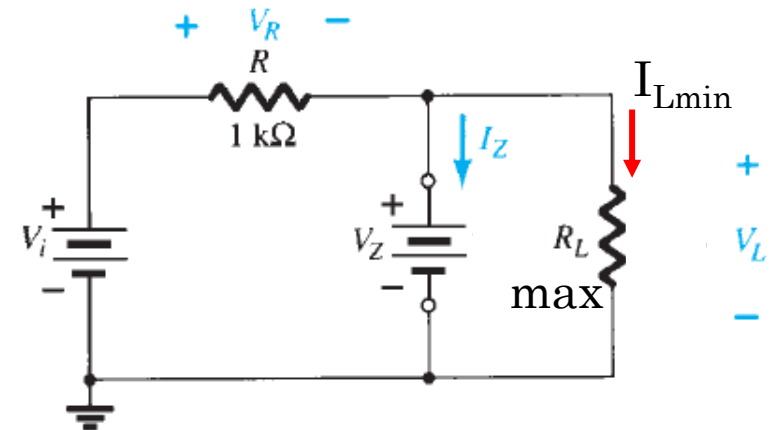
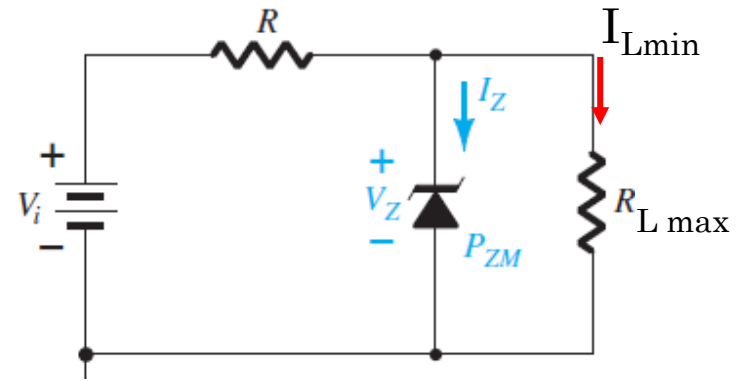
resulting in a minimum I_Z when I_L is a maximum and a maximum I_Z when I_L is a minimum value since I_R is constant.

Since I_Z is limited to I_{ZM} as provided on the data sheet, it does affect the range of R_L and therefore I_L . Substituting I_{ZM} for I_Z establishes the minimum I_L as

$$I_{L_{\min}} = I_R - I_{ZM}$$

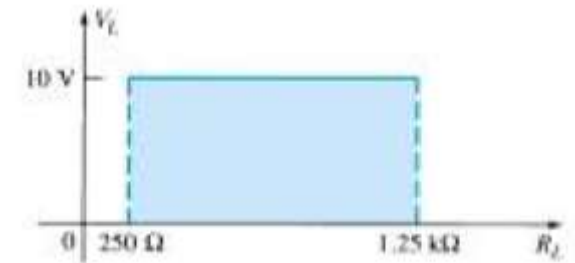
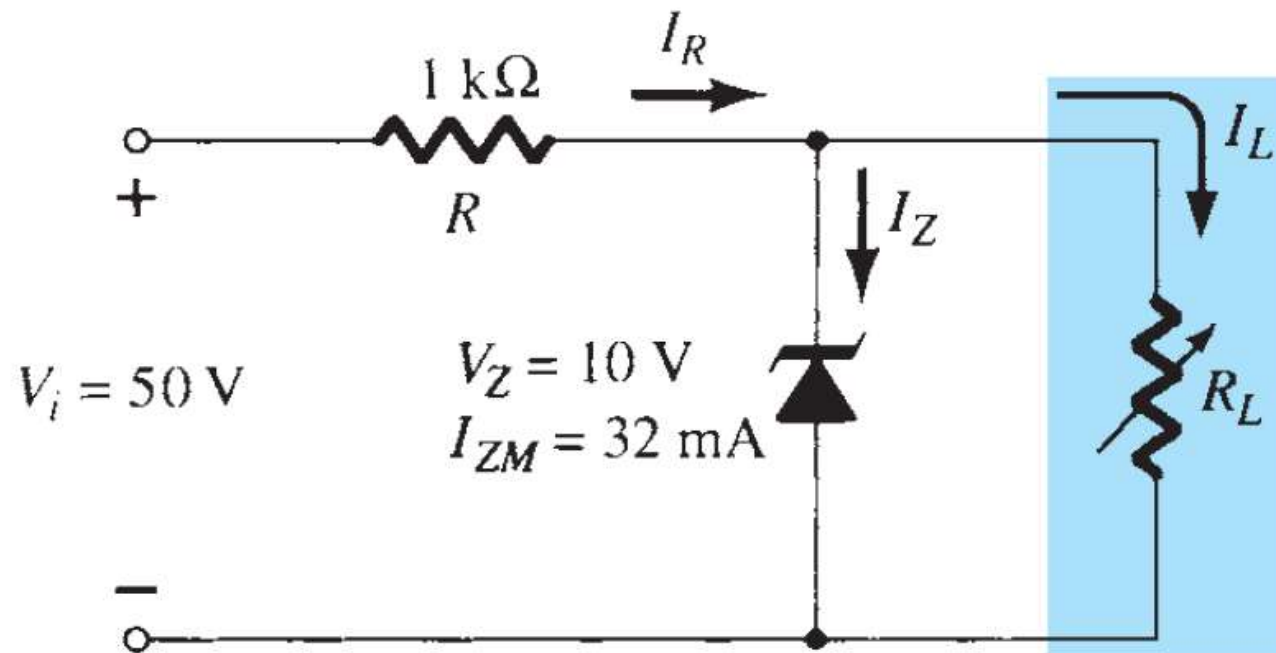
and the maximum load resistance as

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}}$$

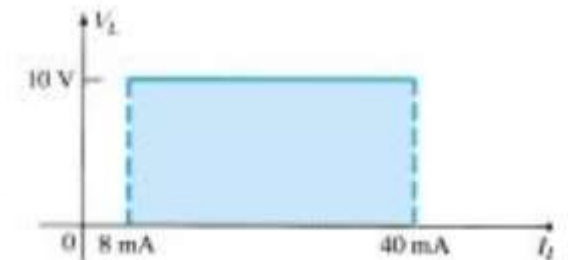


Problem-1

- (a) For the network of Fig. , determine the range of R_L and I_L that will result in V_{R_L} being maintained at 10 V.
- (b) Determine the maximum wattage rating of the diode.

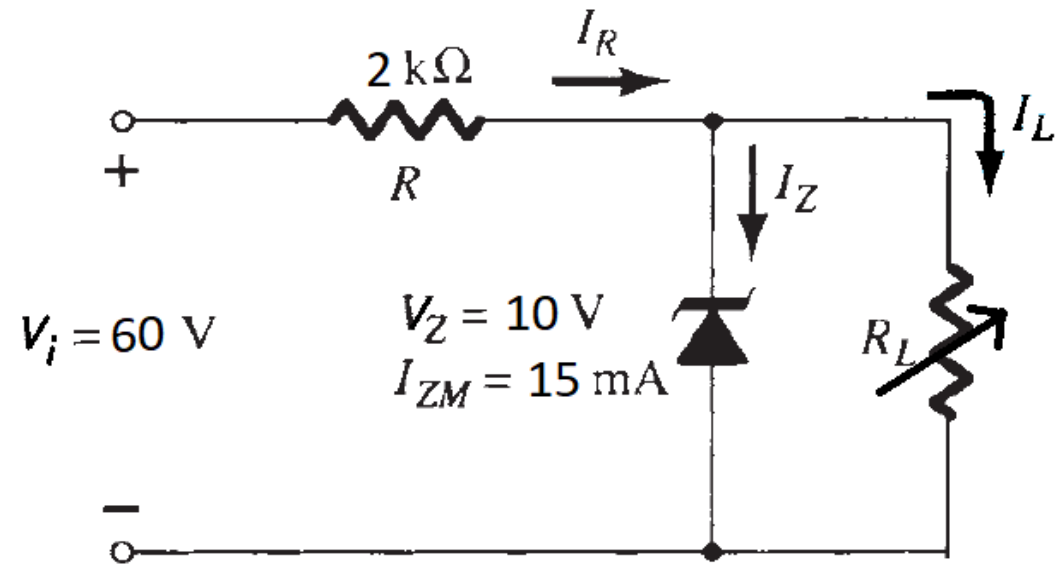


(a)



(b)

2. For the circuit shown in below, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.



Problem-3

A 5.0V stabilized power supply is required to be produced from a 12V DC power supply input source. The maximum power rating P_Z of the Zener diode is 2W.

Using the Zener regulator circuit above calculate:

- a) The maximum current flowing through the Zener diode.
- b) The minimum value of the series resistor, R_S
- c) The load current I_L if a load resistor of $1k\Omega$ is connected across the Zener diode.
- d) The Zener current I_Z at full load.

